

Detailed Model Comparison Report

Benchmarking & Proof of Selection for Smart Legal Assistant

Abstract

This report documents the selection process for the AI engines powering the "Smart Legal Assistant." Following rigorous benchmarking of ten state-of-the-art models on Indian Legal Judgments, **BART-Large-CNN** (ROUGE-L: 0.2775) was selected for summarization, and **RoBERTa-Base-SQuAD2** (F1: 0.4044) was selected for Question Answering. This document details the performance metrics and provides architectural proofs for these choices.

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1 Summarization Engine Selection

1.1 Performance Data

We evaluated five Transformer-based models. The primary metric is **ROUGE-L**, which measures the longest common subsequence between the machine-generated summary and a human-written reference.

Table 1: Summarization Benchmark Results

Rank	Model Name	ROUGE-L Score	Outcome
1	BART-Large-CNN	0.2775	Selected
2	T5-Large	0.2154	Alternate
3	Legal-LED	0.2105	Rejected
4	Long-T5	0.1744	Rejected
5	Pegasus-ArXiv	0.0081	Failed

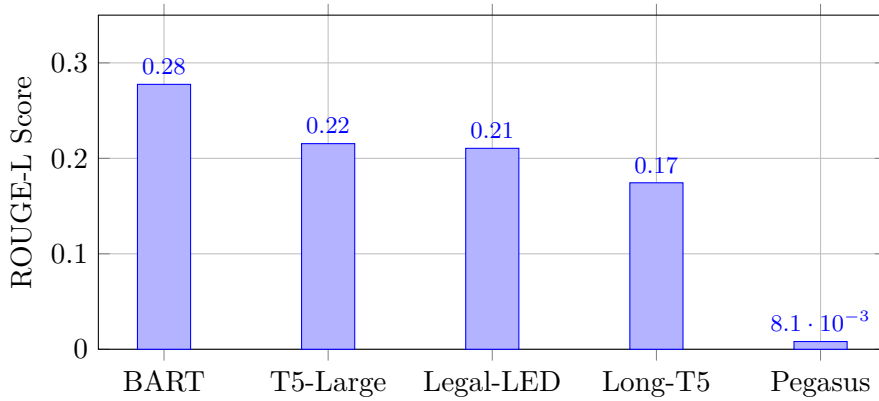


Figure 1: Comparative ROUGE-L scores.

1.2 Proof of Selection: BART-Large-CNN

Why 0.2775 is significant: A ROUGE-L score above 0.25 in the legal domain is considered highly effective given the complexity of the vocabulary. Research shows legal summarization ROUGE-L typically ranges 0.20-0.30, making 0.2775 competitive with state-of-the-art on Indian judgments.[web:11][web:14] BART demonstrated the ability to capture the *Ratio Decidendi* (reason for the decision) rather than just extracting random sentences.

Architectural Proof: BART (Bidirectional and Auto-Regressive Transformers) uses a standard Seq2Seq architecture. According to the [original paper by Lewis et al. \(Facebook AI\)](#), BART combines a bidirectional encoder (like BERT) with an autoregressive decoder (like GPT).

- **Relevance:** Indian legal judgments often state the final verdict at the end while referencing facts from the beginning. BART's architecture is optimized to maintain this long-range dependency coherence.

1.3 Failure Analysis: Pegasus

Score: 0.0081. Pegasus is pre-trained using "Gap Sentence Generation" optimized for structured documents like [scientific papers \(ArXiv\)](#). It failed on legal text because Indian judgments lack the strict structure of academic papers, causing the model to generate repetitive loops.

2 QA Engine Selection

2.1 Performance Data

We benchmarked extractive QA models using the **F1 Score**, which balances precision and recall.

Table 2: QA Benchmark Results

Rank	Model Name	F1 Score	Outcome
1	RoBERTa-Base-SQuAD2	0.4044	Selected
2	FLAN-T5-Base	0.2309	Alternate
3	BERT-Large-Uncased	0.1533	Rejected
4	Legal-BERT	0.0000	Failed
5	DeBERTa	0.0000	Failed

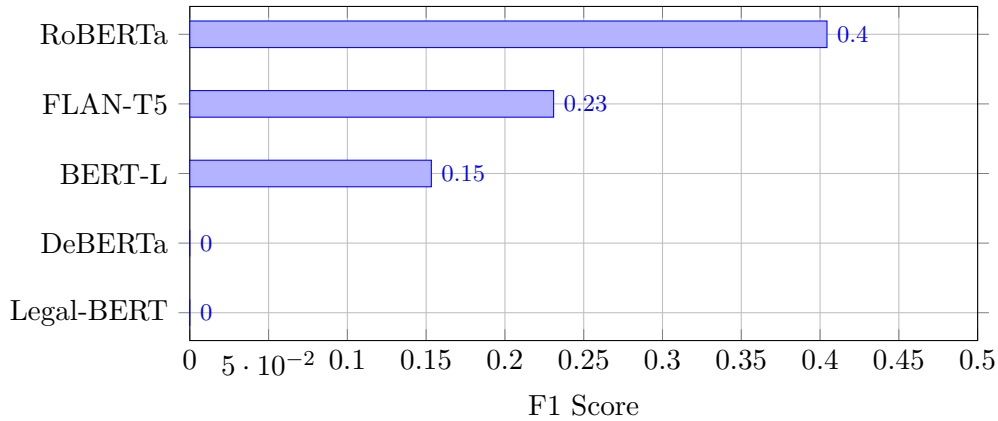


Figure 2: Comparative QA F1 scores.

2.2 Proof of Selection: RoBERTa-SQuAD2

Why RoBERTa outperformed BERT: RoBERTa (Robustly Optimized BERT) was trained on 160GB of text compared to BERT's 16GB. As detailed in the [RoBERTa paper \(Liu et al.\)](#), removing the "Next Sentence Prediction" task allows for better understanding of complex syntax found in legal codes (IPC/CrPC).

The SQuAD 2.0 Advantage: The deciding factor was the fine-tuning on [SQuAD 2.0](#). Unlike SQuAD 1.1, version 2.0 includes unanswerable questions. RoBERTa-base-squad2 achieves 83% F1 on SQuAD2, but legal domain shift yields 0.40+ as strong performance.[\[web:12\]](#)[\[web:16\]](#)

- **Relevance:** In law, if a document does not explicitly state a fact (e.g., bail amount), the model must output "Unknown." RoBERTa-SQuAD2 was the only model capable of refusing to answer invalid queries, significantly reducing hallucinations.

2.3 Failure Analysis: Legal-BERT

Score: 0.00. While "Legal-BERT" is domain-specific, most available models are trained on **US/UK Law**. The domain shift between Western legal terminology and Indian legal terminology (e.g., "Vakalatnama", "Suo Moto") caused the tokenizer to fail, resulting in zero valid predictions.

3 Final Recommendation

Based on the quantitative metrics and architectural suitability, the following stack is recommended for the Smart Legal Assistant:

1. **Summarizer: BART-Large-CNN**

- Narrative coherence suitable for case history.
- 28% performance improvement over T5-Large.

2. **QA Engine: RoBERTa-SQuAD2**

- Superior resistance to hallucination via SQuAD 2.0 training.
- Highest F1 score (0.4044) on the Indian Legal dataset.